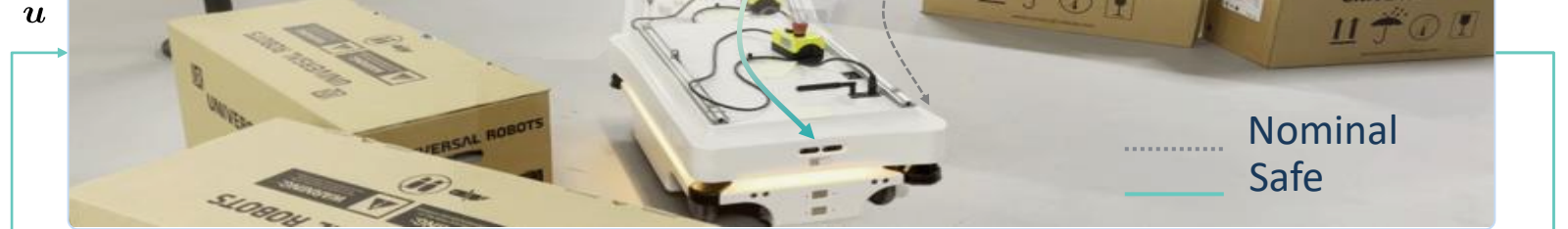


Robot in the environment



Control Input

2) Robust Safety-Critical Control

Robust safety filter (SOCP)

$$\dot{\mathbf{y}}_s = \arg \min_{\dot{\mathbf{y}} \in \mathbb{R}^3} \frac{1}{2} \|\dot{\mathbf{y}} - \dot{\mathbf{y}}_n\|^2$$

s.t. Eq. (13)

$$\nabla \hat{h}(\mathbf{y}, t) \dot{\mathbf{y}} - \epsilon_{\nabla h}(\mathbf{y}, t) \|\dot{\mathbf{y}}\| - C_v \geq -\alpha (\hat{h}(\mathbf{y}, t) - \epsilon_h(\mathbf{y}, t))$$

Low-level velocity controller

$$\mathbf{u} = -K_d(\mathbf{q} - \dot{\mathbf{q}}_s)$$

$$J(\mathbf{q})\dot{\mathbf{q}}_s = \dot{\mathbf{y}}_s \quad J(\mathbf{q}) := \frac{\partial \varphi(\mathbf{q})}{\partial \mathbf{q}} \in \mathbb{R}^{3 \times n}$$

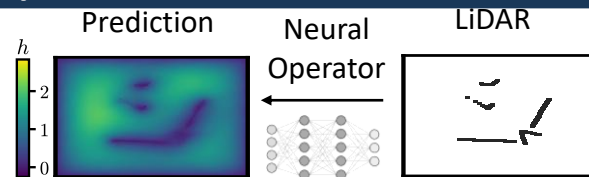
$$\hat{h}(\mathbf{y}, t)$$

$$\nabla \hat{h}(\mathbf{y}, t)$$

$$\epsilon_h(\mathbf{y}, t)$$

$$\epsilon_{\nabla h}(\mathbf{y}, t)$$

1) Fast PSF Inference



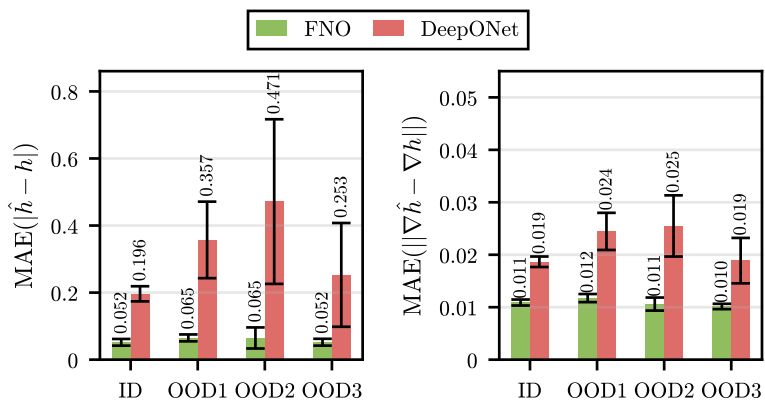
Perception
(e.g. LiDAR)

3) Online Error Bound Estimation

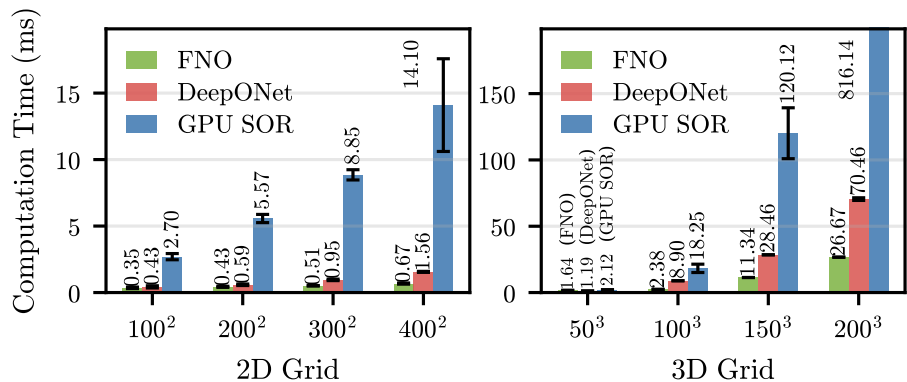
$$\epsilon_h(\mathbf{y}, t) \approx |\hat{e}_h| + \delta_h, \quad \epsilon_{\nabla h}(\mathbf{y}, t) \approx \|\hat{e}_{\nabla h}\| + \delta_{\nabla h},$$

$$\text{Eq. (17), } \min_{\hat{e}_h, \hat{e}_{\nabla h}} \sum_{\mathbf{y}^b \in \mathcal{Y}^b(t)} \left| \underbrace{e_h(\mathbf{y}^b, t) - \hat{e}_h}_{=-\hat{h}(\mathbf{y}^b, t)} - \hat{e}_{\nabla h}^\top (\mathbf{y}^b - \mathbf{y}) \right|^2$$

$\mathbf{y}_k^b$
Obstacle
Boundary



Prediction accuracy



Computational efficiency